

**Investigation into the Theory of Mitigating Telephony Denial of Service Threats to Public
Safety Answering Points Through the Use of Layered Security Practices**



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Author Note

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Abstract

This management investigation report examines the preparedness of Public Safety Answering Points (PSAPs) to detect and mitigate Telephony Denial of Service (TDoS) attacks through the adoption and employment of layered security practices. Using literature review, an interview with an experienced PSAP director, and limited public sentiment surveys, the study compares industry standards from the National Emergency Number Association (NENA) to real-world practices. Findings indicate that while technical solutions and best practices are well documented, implementation across the industry is inconsistent, especially in smaller or underfunded centers. Layered mitigation, combining technical measures with procedural readiness and inter-agency cooperation, offers the greatest operational resilience. Recommendations focus on closing the gap between theory and practice through targeted training, infrastructure modernization, and policy development.

Key words: Public Safety Answering Point (PSAP), Telephony Denial of Service (TDoS), National Emergency Number Association (NENA), layered security.

1.0 Introduction

In the next decade, will PSAPs be able to sustain emergency operations under the growing threat of telephony denial of service attacks? The topic of this management investigation report (MIR) is the preparation of PSAPs to respond to TDoS events with layered security practices. This topic functions on the principles of operational resilience and cybersecurity continuity planning, recognizing that the ability to maintain call-handling capacity during an attack is critical to public safety. The purpose of this report is to evaluate how well PSAPs understand and implement layered defense strategies such as call filtering, geo-fencing, policy based routing, and interagency coordination, compared to published industry standards. Specifically, this MIR has the following objectives:

- Describe the operational need for layered TDoS mitigation in the PSAP environment.
- Determine the degree to which published guidance aligns with the operational realities reported by PSAP leadership.
- Compare published standards and expert perspectives to public perceptions and expectations as assessed through survey.
- Interpret the combined findings to identify gaps in readiness and propose realistic and actionable improvements PSAPs can undertake to mitigate the threat of TDoS and improve their resilience to these attacks.

2.0 Methods of Research

I researched many credible sources to support my analysis of PSAP TDoS preparedness. I began my research in July 2025, reviewing literature on TDoS mitigation, NG9-1-1 infrastructure, layered defense strategies, and real-world instances of PSAPs impacted by TDoS attacks. Sources were located using the Columbia Basin College Library's EBSCO Discovery Service, as

well as targeted searches of NENA, CISA, and other professional and government publications, and historical reporting on documented TDoS attacks impacting PSAPs in the United States. Most sources are current from 2015 – 2025, with A comprehensive study of flooding attack consequences and countermeasures in Session Initiation Protocol (Hussain, et. al. 2015) being the oldest. Two of the most valuable sources are NENA-INF-011.2-2020, which provides a detailed framework of the employment of policy routing rules, and CISA’s 2022 case study, which offers lessons learned from an actual TDoS event that impacted a PSAP. In addition to the literature review, I conducted a structured interview with Tina Buneta, ENP, CPE, RPL, Director of Aurora911, using seven prepared questions. The questions were provided early, and the interview was conducted remotely via video call. I also created an original survey to gauge public perception of TDoS risks and potential mitigation strategies, which was completed by 14 respondents enrolled in various Columbia Basin College Bachelors Programs. Data from these methods were compared to identify consistencies, conflicts, and actionable insights.

3.0 Findings and Interpretations

The following sections integrate findings from three data sources- prior published research, an expert interview, and an original survey. These sources are used together to assess how layered mitigation strategies are understood and implemented in PSAP operations. By comparing theoretical frameworks to operational realities and public expectation, this analysis identifies strengths, weaknesses, and opportunities for improving emergency communications resilience.

3.1 Theory and Research Summary

The topic of my management investigation report (MIR) is the preparation of Public Safety Answering Points (PSAPs) to respond to Telephony Denial of Service (TDoS) attacks. This topic functions on the principles of operational resilience and cybersecurity continuity

planning. TDoS events, like robocall floods, SIP based attacks, or automated 911 dialing exploitation can overwhelm critical emergency communications systems and prevent legitimate calls from being received. As documented in my annotated bibliography and recent events, TDoS mitigation remains inconsistent across agencies (NENA, 2024; Hussain et al., 2015; Hosseinpour et al., 2018), leaving many PSAPs reliant primarily on telecom providers for defense (Cybersecurity and Infrastructure Security Agency [CISA], 2022); National Emergency Number Association [NENA], 2024; Hussain et al., 2015). These sources examine various dimensions of the threat, including real-world incidents, detection models, technical countermeasures, and the importance of formalized staff protocols. Research in this area is crucial because PSAPs represent the first link in the emergency response chain. Their failure to receive or respond to calls during a TDoS attack may result in delays that put lives at risk. This review of sources informs my MIR by identifying practical strategies and highlighting areas in which current PSAP protocols fall short.

Definitions of Key Terms and Concepts

For the purposes of this review, the following terms and ideas will be relied upon and are explained briefly here. (MS-ISAC, 2017) clearly defines a TDoS attack as a deliberate exhaustion of telephone resources, preventing incoming or outgoing calls, often targeting PSAPs and government agencies. (NENA, 2025) specifically discusses call interrogation procedures and the generation of a call suspicion score in the context of TDoS mitigation. Caller interrogation is the automated process of gathering information about the caller and caller's device, such as its location, call history, device type, etc. The suspicion score is an assessment of the likelihood of an incoming call not being genuine based on the results of the interrogation. (Hussain et al., 2015) defines Session Initiation Protocol (SIP) as a technology that transmits phone calls over

VoIP based networks. In the context of emergency communications, Next Generation 9-1-1 (NG9-1-1) refers to a modern, IP-based infrastructure that enables voice, video, and text communications to be routed to appropriate PSAPs (NENA, 2024). Another important term, policy routing rules, describes a system for directing emergency calls based on logic that considers network conditions, PSAP status, and caller location (NENA, 2020). These definitions frame my analysis of layered TDoS defenses and how organizations assess and ensure resilience in the modern emergency communications environment.

Conceptual Model

A key conceptual model discussed in the standards outlined in NENA-INF-011.2-2020 explains how policy routing rules can be adapted to mitigate a TDoS attack (NENA, 2020). The model illustrates how calls can be prioritized, rerouted, or dropped based on call interrogation results including the type of device placing the call, location of the device, and history of the device and others in the same area. It further allows a query of the status of the primary responsible PSAP as well as available backup PSAPs in the area that could handle the call. This enables greater survivability during mass call events. For example, when one PSAP becomes overwhelmed or compromised, the routing system can dynamically reassign traffic to functioning centers or redirect to non-emergency lines. As NENA (2020) emphasizes, the success of such a model depends not only on technical infrastructure, but on pre-defined policy logic, collaboration with neighboring agencies, and continuous monitoring of system status. Figure 4 demonstrates how these concepts work together to screen and direct calls while monitoring network status.

Conceptual Model of Policy Routing Rules

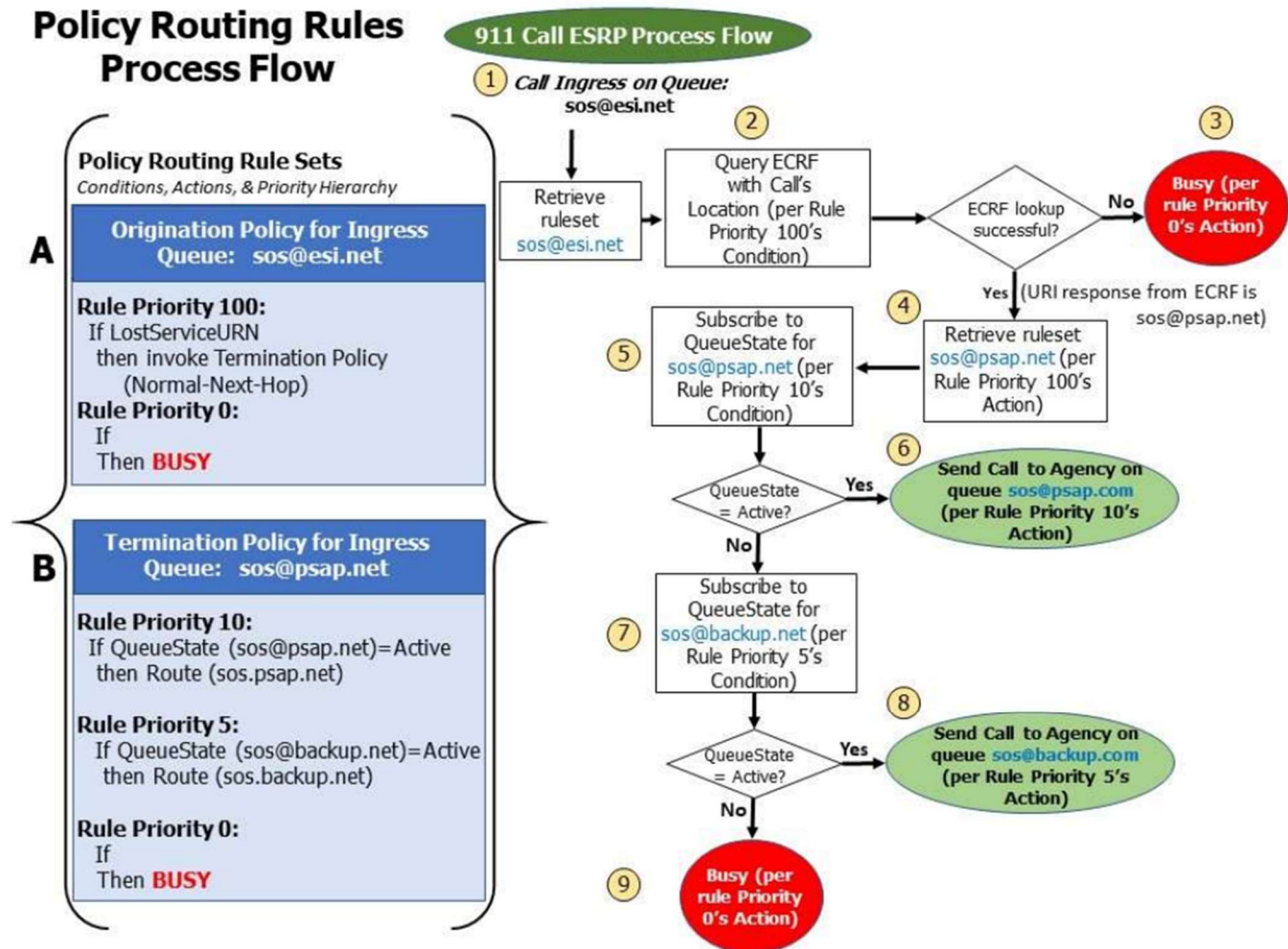


Figure 1. The intended flow of an inbound call via Policy Routing Rules (NENA, 2020).

According to Figure 1 (NENA, 2020), when a 911 call is received, it's placed into a specific queue, in this case represented as sos@esi.net. This is step 1 in the model. The agency's policy routing rules set, a predetermined set of values to query, and outcomes for different call routing events then take over. In step 2, the caller is queried to establish good location data. This can come from a GPS plot of the device's location, cellular data triangulation, or a physical address based on VoIP routing data or a preprogrammed VoIP address value. At step 3, if no location can be determined, the PRR's have a chance to refuse the call and transfer it to a busy signal queue

until the caller disconnects. This is often where spoofed VoIP calls or call flooding attacks can first be identified. If the location data a PSAP is receiving doesn't make sense, or is obfuscated, that's a potential indicator of malicious calling. If the call is legitimate then steps 4 through 7 deal with regular call handling, to assign the incoming call to the correct queue for handling. However, agencies with effective PRRs can also minimize the impact of a TDoS attack at step 7. If an agency falls victim to call flooding and is no longer able to accept incoming calls to their queue, correctly set PRRs can read the queue status as unavailable and automatically redirect subsequent legitimate calls to other agencies predefined in the PRRs as appropriate. Thus, the model shows how a successful implementation of policy routing rules can potentially mitigate the chances of a TDoS attack crippling a PSAP but also improve the ability for 911 calls to be answered even if a PSAP does experience a TDoS event.

Important Findings

Multiple sources reveal findings that demonstrate the real-world impacts of TDoS threats. For example, CISA (2022) recounts a prolonged TDoS attack against a PSAP's non-emergency lines that forced the agency to collaborate with telecom providers and neighboring jurisdictions without prior planning, exposing the lack of staff training and formal procedures prior to the event. Likewise, Khandelwal (2016) reports on a TDoS incident caused by a JavaScript exploit that caused iPhones to flood PSAPs with 911 calls, again showing how simple, low-resource attacks can have widespread effects. On the technical side, Hussain et al. (2015) and Hosseinpour et al. (2018) highlight anomaly-based detection strategies that identify unusual patterns of telephone traffic, which can signal an attack. For example, monitoring the frequency and volume of SIP messages (used for setting up VoIP phone calls) can quickly detect a sudden spike in calls indicative of an attack. Similarly, bloom filters—a method of quickly determining

whether incoming calls are likely legitimate—can rapidly screen out falsified caller IDs (known as spoofing), thus preventing malicious calls from overwhelming emergency phone lines.

Despite the availability of these strategies, recent real-world incidents continue to demonstrate significant vulnerabilities in emergency call centers, underscoring an ongoing need for PSAPs to have clear, practiced plans to respond effectively to TDoS attacks (CISA, 2024). Versadial Solutions (2024) emphasizes that even short-duration attacks can significantly disrupt critical PSAP services. Together, these findings underscore the necessity of both proactive planning and technical adaptability to ensure continuity of operations in emergency response.

Research Trends

Several trends are shaping how PSAPs are approaching TDoS defense. The first is a shift from legacy telephony to SIP/VoIP-based NG9-1-1 systems, which brings increased flexibility, but also greater cyber exposure (Hosseinpour et al., 2018; NENA, 2024), especially from the emerging trend of Internet connected devices (IoT Devices) that are starting to become vulnerable to cyber threats (NENA, 2025). A second notable trend is the increased adoption of layered defense strategies by local emergency communication centers. Rather than relying solely on protection provided by telephone carriers (e.g., basic call blocking from phone companies), local agencies are now encouraged to establish their own detailed procedures (Standard Operating Procedures or SOPs) to handle potential attacks. For instance, agencies might implement staff training programs to help call-takers recognize early signs of an attack, such as an abnormal influx of calls or calls from suspicious phone numbers. Additionally, agencies may employ network monitoring tools—software that continuously observes call traffic—to detect and quickly respond to unusual call patterns that could signal a TDoS event (CISA, 2022;

NENA, 2024). A third trend is the growing reliance on collaborative threat intelligence, where PSAPs work hand in hand with vendors, regional partners, and federal agencies to identify and respond to cyber threats (CISA, 2022). These patterns suggest that while the threat landscape is evolving, so too are the available mitigation strategies. Unfortunately, not all PSAPs are equally prepared to adopt sufficient defenses.

Conflicts of Ideas or Approaches

Despite general agreement, some conflicts exist within the sources I've reviewed regarding the priority of implementation. For instance, NENA (2024) stresses that PSAPs must assume internal responsibility for developing internal TDoS response plans, whereas CISA (2022) documents that many centers remain reliant on telecom providers and informal practices, not meeting the expectations set by NENA. Similarly, while technical detection strategies such as anomaly detection and automated call filtering have been proven effective in controlled laboratory conditions, there is very limited research demonstrating how effectively these tools work in real-life Public Safety Answering Point (PSAP) operations. For example, techniques that rely on precise traffic analysis to detect unusual call volumes may struggle when legitimate call traffic unexpectedly surges, such as during natural disasters or public emergencies.

Consequently, there is still uncertainty about whether these advanced technical solutions, which function reliably in ideal lab conditions, can adapt effectively to unpredictable, high-stakes scenarios common in actual emergency call centers. These conflicts raise questions about the practicality, cost, and training burdens associated with more sophisticated solutions, particularly for rural or underfunded centers.

Research Conclusions

Based on the findings of my research, I can conclude the following:

1. A majority of my sources agree that PSAPs need a layered defense approach that includes both technical and operational components.
2. TDoS attacks are increasingly feasible due to modern VoIP vulnerabilities and require more than just carrier level protection.
3. There is a need to verify whether PSAPs understand and can reasonably apply a conceptual model like the policy routing rules standard set forth by NENA, or if they remain theoretical in practice with my interviewee.

Gap in Knowledge or Research

While high-level guidelines are prevalent, practical implementation remains unclear or unattainable for many local PSAPs (SecureLogix, 2020). For instance, NENA (2024) lays out broad security standards but stops short of examining how PSAPs execute them on the ground. The aim of my MIR is to explore whether these standards are understood and implemented at the local level, and to what extent organizational culture, training, and collaboration affect real-world resilience. This fills a critical knowledge gap and may help public safety stakeholders better allocate resources and training to improve emergency response continuity.

3.2 Interview Results

Informant Profile

Tina Buneta, ENP, CPE, RPL, currently serves as the Director of Aurora911. In this capacity, she oversees dispatchers, supervises call-handling operations, and coordinates with city IT, telecom carriers, and regional PSAP partners to ensure successful operations around the clock. As the Director of Aurora911, Tina has worked to ensure Aurora911 is

ready and capable of responding to both routine call surges and major disruption events, giving her direct insight into the operational and organizational challenges of mitigating a Telephony Denial of Service (TDoS) attack. (T. Buneta, personal communication, 07/24/2025).

Inductive Analysis

1. Context and conditions

According to T. Buneta (personal communication, 07/24/2025), TDoS attacks remain a relatively new consideration for many PSAPs. Mitigation is typically seen as the responsibility of cyber/IT teams rather than operational staff, which limits PSAP involvement in proactive planning. The interviewee noted that most U.S. PSAPs have four or fewer call-taking positions and often compete with police and fire for scarce public safety funding. Additionally, operational siloing, where responsibilities are divided across departments, can often reduce visibility and direct control over response measures. For example, if a cities IT department manages network monitoring and security, they might not independently prioritize the enhancement of layered security measures for the cities PSAP, like the PSAPS director and operations team would. Thus, PSAP staff might find it hard to secure funding or support for the IT department to buy in on investing in new capabilities or improved protections.

2. Causal Relationships

The interviewee emphasized that delayed recognition of a TDoS event often stems from its initial similarity to normal call spikes caused by mass casualty incidents, high visibility accidents, or public disturbances. Because telecom providers can take

hours to block malicious sources, late detection directly increases service degradation. Furthermore, resource disparities between PSAPs, often driven by local tax base limitations, create uneven adoption of modern mitigation tools (T. Buneta, personal communication, 07/24/2025). For instance, larger urban centers may implement AI-assisted call pattern analysis or advanced SIP traffic filtering, while smaller rural PSAPs may struggle to employ even a simple layered defense strategy, choosing instead to rely solely on basic number blocking features provided by their telecom providers.

3. Actions / Interactions

When unusual call volumes occur, the interviewee described initiating immediate notifications to supervisors, engaging telecom providers, and coordinating with neighboring PSAPs. While NG9-1-1 capabilities and policy-based routing are recognized as best practices, most agencies apply them only superficially due to cost, the existence of legacy infrastructure, and limited expertise. Some agencies are beginning to explore AI-assisted call pattern analysis, but widespread use remains rare (T. Buneta, personal communication, 07/24/2025).

4. Intended and Unintended Consequences

Layered defenses such as geo-fencing, IP or device blacklisting, and filtering of malicious calls to automated attendants can reduce load and preserve 911 capacity. However, these measures risk blocking legitimate calls, a consequence PSAPs are often unwilling to accept. The interviewee noted that balancing protection with access is a constant challenge, and fear of inadvertently rejecting real emergency

calls often delays implementation of aggressive filtering measures (T. Buneta, personal communication, 07/24/2025).

Deductive Analysis

The Policy Routing Rule conceptual model found in section 3.1 Review of Sources highlights how policy-based routing, geospatial logic and layered defenses, including call suspicion scoring, automated rerouting, and inter-PSAP cooperation are all essential to continuity of operations during an event like a TDoS attack where a PSAP becomes unreachable. The model assumes proactive detection, rapid call interrogation and routing decisions, and close coordination with telecom and regional PSAP partners.

Model vs. Informants' Answers

Informant's answer = elements of model

- The interviewee affirmed that inter-agency coordination, layered mitigation, and filtering of malicious calls align directly with the model's principles. She noted these approaches reflect the intent of her own PSAP and others and are considered best practice (T. Buneta, personal communication, 07/24/2025).

Informant's answer $\pm$ *close to* but slightly different than model

- While the model envisions automated, policy driven routing, the interviewee described a process that remains heavily reliant on manual recognition and decision making, with limited automation currently deployed.

Informant's answer $\neq$ contradicts one or more parts of model

- The model assumes consistent capability across PSAPs, but the interviewee stressed that resource disparities, particularly in small, rural centers, makes full implementation of

features like geo-fencing, or AI based detection impractical in many jurisdictions (T. Buneta, personal communication, 07/24/2025).

Based on this interview, I estimate that the interviewee's operational experience aligns with approximately 75-80% of the conceptual model. While the core tenets of layered defenses and cooperative and policy-based routing are present, the envisioned automation and uniform adoption across the industry are significantly hindered by limited funding, legacy systems, and organizational silos.

Summative Conclusions

From the inductive and deductive analyses of this interview, two generalizations are apparent:

1. Effective TDoS mitigation in PSAPs depends as much on organizational culture, training, and inter-agency relationships as on technical knowledge and safeguards.
2. Although a layered defense strategy is widely supported in theory, real world deployment is constrained by limited resources, infrastructure limitations, and risk aversion towards inadvertently blocking legitimate 911 calls.

3.3 Student Survey Results

Survey Process

I designed a public facing survey intended to gauge public perceptions of PSAP exposure to TDoS and the practical value of layered mitigations (e.g., call filtering, geo filtering, policy-based routing, SOPs/training, interagency rerouting of calls). The survey mixed Likert, multiple choice, yes-no, and ranked order questions, as well as short answer prompts. I analyzed 14 written responses, tallied the responses manually, and created three graphs of note using Excel.

Inductive Analysis

The following graphs represent the most important results from my original survey. The clustered column graph represents public desire for local PSAP involvement in cybersecurity planning/response; the bar chart shows the public's perceived effectiveness of potential countermeasures to TDoS attacks; and the donut chart compares the public's comfort with adoption of aggressive mitigation strategies like blocking inbound calls, even if there's a risk of blocking a legitimate caller.

**Figure 2 – Belief That Local PSAPs Should Have More Direct Involvement In
Cybersecurity Planning**

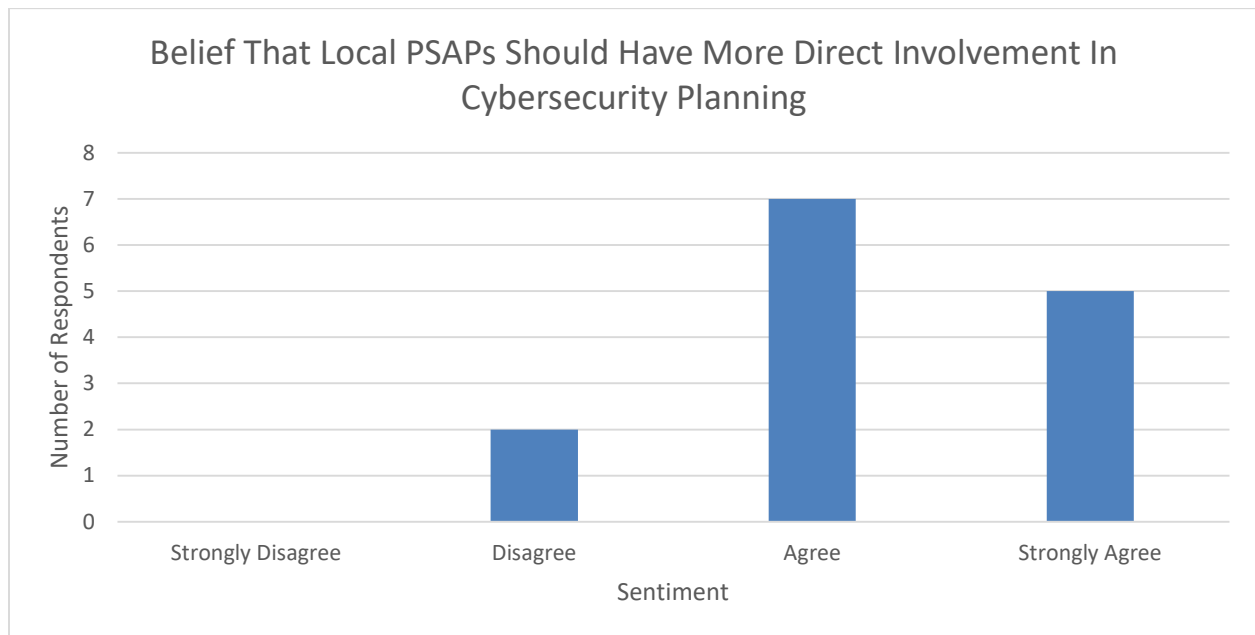
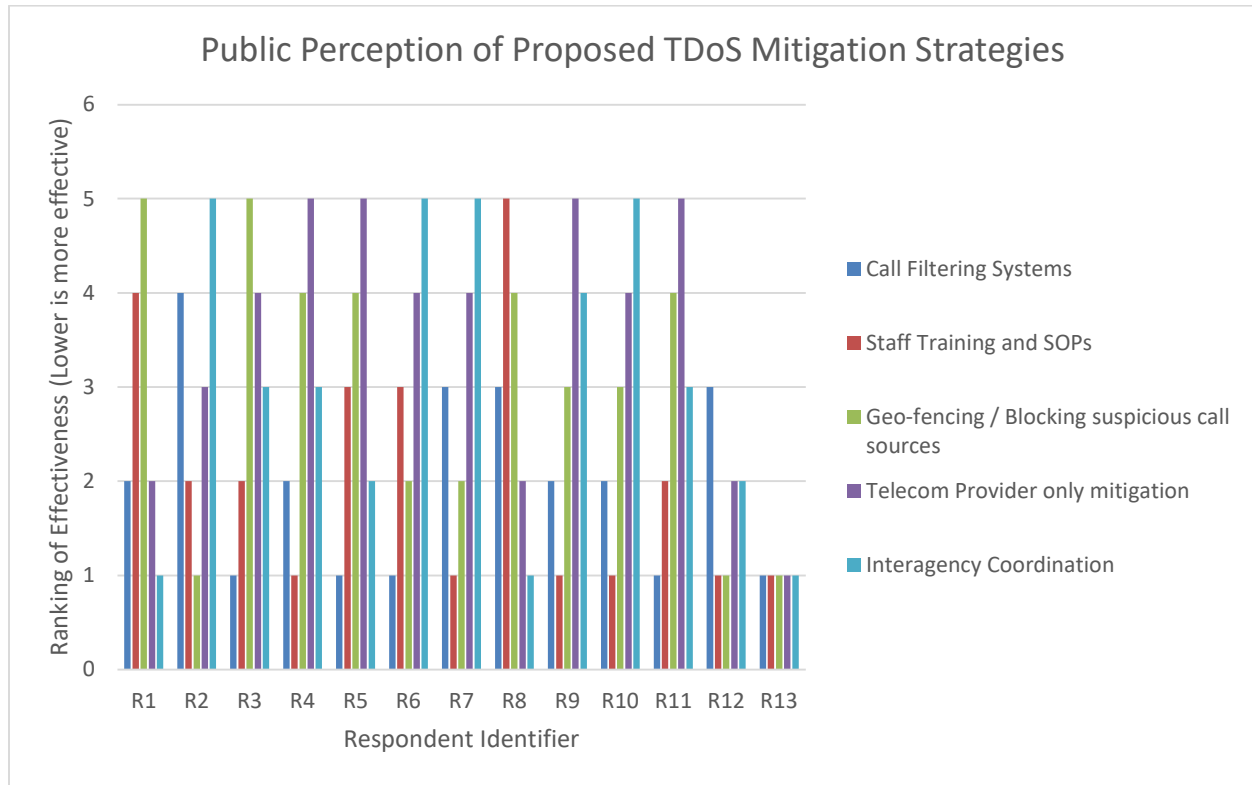


Figure 2. 12 of 14 (86%) endorse direct PSAP involvement in cybersecurity planning/response.

Figure 3 – Public Perception of Proposed TDoS Mitigation Strategies

Please note: Three respondents treated this question as per-item ratings instead of a unique 1-5 ranking. I included their responses in the interest of completeness.

Figure 3. Among the layered security strategies proposed in the survey, respondents reported call filtering as the most effective, while telecom protections and interagency coordination lagged.

**Figure 4 – Willingness to Block Suspicious Incoming Calls to 9-1-1, Even if Potential Exists
to Block Genuine Calls**

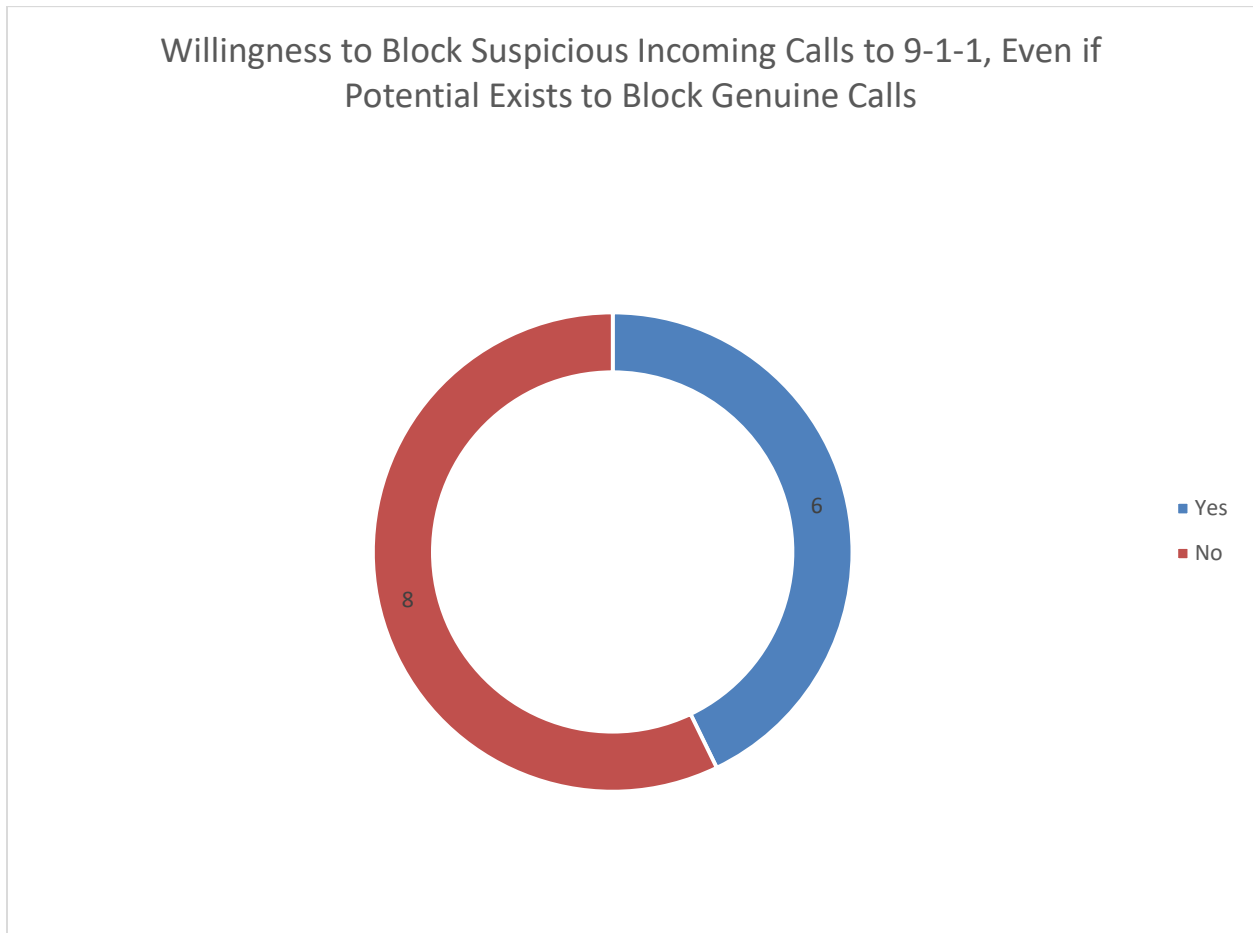


Figure 4. Comfort with temporary blocking is split: 8 No (57%) vs 6 Yes (43%).

Deductive Analysis

1. The Policy Routing Rule conceptual model found in section 3.1 Review of Sources highlights how policy-based routing, geospatial logic and layered defenses, including call suspicion scoring, automated rerouting, and inter-PSAP cooperation are all essential to continuity of operations during an event like a TDoS attack where a PSAP becomes unreachable. The model assumes proactive detection, rapid call interrogation and routing decisions, and close coordination with telecom and regional PSAP partners.

2.

Survey results = elements of model

There is strong public endorsement of local PSAP involvement and a higher perceived value in the use of filtering and improved SOPs to match the conceptual model's guidance on PSAP owned layered mitigation strategies.

Survey results $\pm$ *close to* but slightly different than model

The support for policy-based routing mitigations appears lower than support for GEO-fencing filtering and the improvement of SOPs, suggesting that either awareness or implementation of the measures demonstrated in the conceptual model lag NENA's preferred implementation rate.

Survey results $\neq$ contradicts one or more parts of model

The use of telecom carrier only protection ranks lowest, contradicting any assumption that upstream defenses or siloed reliance on other agencies is an adequate mitigation strategy for TDoS threats.

3. Based on my English 410 survey results, I estimate that the respondents closely agree (~85-90%) with the layered model illustrated by the NENA conceptual model; however, responses about policy-based routing mitigation and willingness to block malicious calls indicates uneven support that should be addressed.

Summative Conclusions

1. The awareness of TDoS risk is relatively high with both the public and industry leaders, but implementation is lopsided. More basic call filtering is common, while policy-based routing controls and formal SOPs lag the curve.

To close the gap, PSAP leaders should focus on advocating for and owning the operational processes-where able- to implement a layered defensive strategy. This should include updated staff protocols and training, updated technological support and screening, refreshed infrastructure, and a focus on inter-agency rerouting agreements, rather than relying on telecom provider level protections alone.

4.0 Conclusions

Across all three research methods, a consistent pattern emerges; Layered mitigation strategies offer the strongest protection against TDoS threats, but their adoption is uneven. Literature and industry standards clearly emphasize the need for PSAP-owned security measures such as automated call interrogation, suspicion scoring, geo-fencing, and policy-based routing rules in addition to carrier-based protections. The expert interview confirmed that while many PSAP leaders support these measures in principle, real-world deployment is often hindered by funding limitations, legacy infrastructure, and operational siloing. Survey results suggest the public values proactive PSAP involvement in cybersecurity and support measures like call filtering, though opinions are divided on aggressive blocking that risks excluding legitimate calls. The

most important lesson is that both technical readiness and organizational culture are critical. Even the most advanced systems require trained personnel, clear SOPs, and inter-agency agreements to be effective during a live attack.

5.0 Recommendation

PSAP managers, whether new or experienced, should treat layered TDoS mitigation as a core component of operational continuity planning, not as an optional enhancement. Based on this study's findings, the following actions are recommended:

1. Adopt a formal layered defense plan incorporating NENA's policy routing model, call filtering, geo-fencing, and inter-agency rerouting protocols.
2. Invest in staff training and simulation exercises so call-takers and supervisors can recognize early indicators of a TDoS event and respond according to predefined SOPs.
3. Collaborate with telecom providers and regional partners to ensure mutual aid agreements and routing logic are in place before an incident occurs.
4. Modernize infrastructure incrementally, prioritizing capabilities that enable automated call interrogation and rapid rerouting.
5. Engage in public communication to balance transparency about security measures with the need to maintain trust in emergency services.

Further research is justified to measure the real-world effectiveness of advanced detection and filtering tools in live PSAP environments. The framework outlined here could also serve as the basis for a BAS capstone or master's thesis, focusing on operationalizing TDoS mitigation strategies across agencies of varying sizes and resources.

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